

Text Document

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System Architecture Design

Overview

The Flood Prediction System follows a machine learning-based architecture that combines data processing, model training, and a web-based prediction interface. The system collects environmental and geographical data, processes the information, applies machine learning algorithms, and generates flood risk predictions.

Architecture Components

1. User Interface Layer

The user interacts with the system through a Flask-based web application. The interface allows users to enter environmental parameters required for prediction.

Input parameters include:

- Latitude
- Longitude
- Rainfall
- Elevation
- Slope
- Distance from water bodies
- Flood duration

2. Application Layer

The Flask application acts as a bridge between the user interface and the machine learning model.

Responsibilities:

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- Receive user input
- Validate input data
- Send processed data to the ML model
- Display prediction results

3. Data Processing Layer

The input data is processed before prediction.

Processing steps:

- Data cleaning
- Handling missing values
- Feature selection
- Data transformation
- Normalization

4. Machine Learning Layer

The trained machine learning model analyzes input parameters and predicts flood risk.

Algorithms considered:

- Decision Tree
- Random Forest
- K-Nearest Neighbor (KNN)
- XGBoost

5. Output Layer

The final prediction is displayed to the user through the web interface.

Output:

- Flood risk level
- Prediction result

- Risk information

System Architecture Flow

User

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Web Interface (Flask)

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Data Processing

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Machine Learning Model

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Flood Risk Prediction

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Display Result